

## Building Models

The framework of Bayesian networks is a very efficient language for building models of domains with inherent uncertainty. However, as can be seen from the calculations in Section 2.6, it is a tedious job to perform evidence transmission even for very simple Bayesian networks. Fortunately, software tools that can do the calculation job for us are available. In the rest of this book, we assume that the reader has access to such a system (some URLs are given in the Preface). Therefore, we can start by concentrating on how to use Bayesian networks in model building and defer a presentation of methods for probability updating to Chapter 4.

In Section 3.1, we examine through examples the considerations you may go through when determining the structure of a Bayesian network model. Section 3.2 gives examples of estimation of conditional probabilities. The examples cover theoretically well-founded probabilities as well as probabilities taken from databases and purely subjective estimates. Section 3.3 introduces various modeling tricks to use when the amount of numbers to acquire is overwhelming. Finally, Section 3.4 considers other types of queries that can be answered by Bayesian networks besides standard probability updating.

### 3.1 Catching the Structure

The first thing to have in mind when organizing a Bayesian network model is that its purpose is to give estimates of certainties for events that are not directly observable (or only observable at an unacceptable cost), and the primary task in model building is to identify these events. We call them *hypothesis events*. The hypothesis events detected are then grouped into sets of mutually exclusive and exhaustive events to form *hypothesis variables*.

The next thing to have in mind is that in order to come up with a certainty estimate, we should provide some information channels, and the task is to identify the types of achievable information that may reveal something about the hypothesis variables. These types of information are grouped into

*information variables*, and a typical piece of information is a statement that a certain variable is in a particular state, but softer statements are also allowed.

Having identified the variables for the model, the next thing will be to establish the directed links for a causal network.

### 3.1.1 Milk Test

Milk from a cow may be infected. To detect whether the milk is infected, you have a test, which may either give a *positive* or a *negative* test result. The test is not perfect. It may give a positive result on clean milk as well as a negative result on infected milk.

We have two hypothesis events: *milk infected* and *milk not infected*, and because they are mutually exclusive and exhaustive, they are grouped into the variable *Infected?* with the states *yes* and *no*. A possible information source is the test results, which can be either *positive* or *negative*. For this, we establish the variable *Test* with states *pos* and *neg*.

The causal direction between the two variables is from *Infected?* to *Test* (see Fig. 3.1).



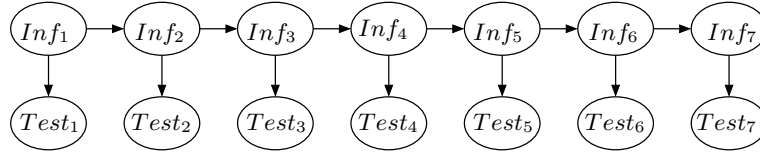
**Fig. 3.1.** The Bayesian network for the milk test.

**Warning:** Certainly, no sensible person will claim that a positive test result may infect the milk. However, our reasoning is often performed in the diagnostic direction, and in more complex situations you may therefore be tempted to wrongly direct the link from “symptom” to “disease”.

From one day to another, the state of the milk can change. Cows with infected milk will heal over time, and a clean cow has a risk of having infected milk the next day. Now, imagine that the farmer performs the test each day. After a week, he has not only the current test result but also the six previous test results. For each day, we have a model like the one in Fig. 3.1. These seven models should be connected such that past knowledge can be used for the current conclusion. A natural way would be to let the state of the milk yesterday have an impact on the state today. This yields the model in Figure 3.2.

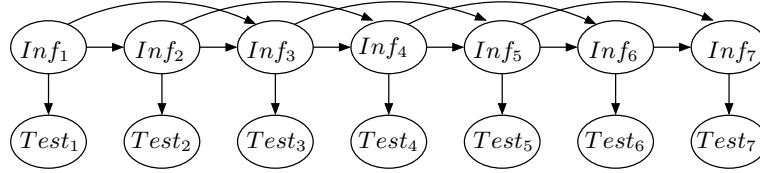
The model in Figure 3.2 contains a set of hidden assumptions, which can be read from the d-separation properties.

First, the model assumes the *Markov property*: if we know the present, then the past has no influence on the future. In the language of d-separation, the assumption is that, for example,  $Inf_{i-1}$  is d-separated from  $Inf_{i+1}$  given  $Inf_i$ . If we know that the milk on day four is infected, then this can be used



**Fig. 3.2.** A seven-day model for milk test.

to forecast the probability that the milk will be infected on day five. This forecast will not be improved by knowing that the milk was not infected on day three. For various diseases, such an assumption will not be valid. Some diseases have a natural span of time. For example, if I have the flu today but was healthy yesterday, then I will most probably have the flu the day after tomorrow. On the other hand, if I have had the flu for four days, then there is a good chance that I will be cured the day after tomorrow. If the Markov property of Figure 3.2 does not reflect reality, the model should be changed. For example, it may be argued that you also need to go an extra day back, and the model will be as in Figure 3.3.

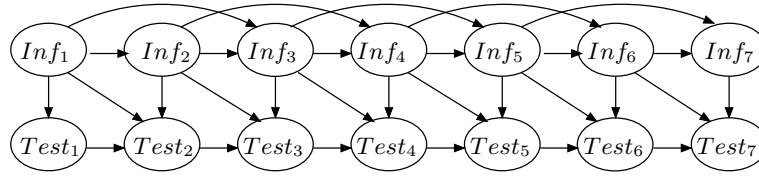


**Fig. 3.3.** A seven-day model with a two-day memory of infection.

Notice that although we in practice will never know the state of the infection nodes, it makes a difference whether the memory links are included. In the reasoning, we cannot exploit knowledge of the exact state of the previous infection node, but we may use a probability distribution based on a test result.

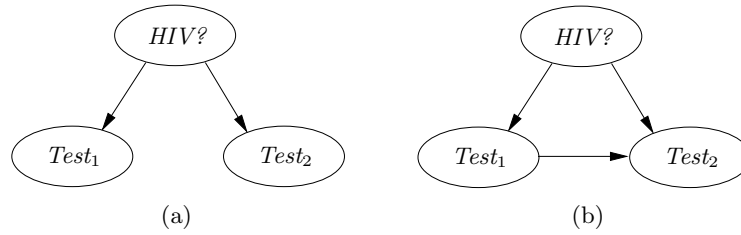
The second hidden assumption has to do with the test. Any two test nodes are d-separated given any infection node on the path. This means that the fault probability of the test is independent of whether it was previously correct. In other words, the fact that the test was wrong yesterday has no influence on whether the test will be correct today. If this does not reflect the behavior of the test, you may, for example, include its performance yesterday in the model. This is done in Figure 3.4.

**A minor digression on modeling of tests:** It is good to have as a rule that no test is perfect. Unless you explicitly know otherwise, a test should always be given a positive probability of false positives as well as false negatives. This is not all, though. You should also take the mechanism for false test



**Fig. 3.4.** A seven-day model with two-day memory for infection and a one-day memory of correctness of test.

results into account. Consider for example an HIV test with a probability of false positives of  $10^{-5}$ , and assume that a person has received a positive test result. Now, you may have the option of repeating the test, but will this be of any help? It will depend on the mechanism that cause the test to give a wrong result. If a test is positive because this particular person's blood is composed s.t. it will produce a positive test result regardless of a positive HIV infection, then a repeated test will not provide new information. If, on the other hand, the experiment is so that it now and then goes wrong, then a repeated test may be worthwhile and it will be advisable to repeat the test before the "verdict" is passed (in case the second test result is negative, a third test may be advisable). Models for these two types of failure mechanisms are shown in Fig. 3.5; the models do not mix the two types of failure.



**Fig. 3.5.** Model (a) illustrates the scenario where a repeated test may provide new information, and model (b) shows the situation where repeating a test always produces the same result.

### 3.1.2 Cold or Angina?

I wake up in the morning with a sore throat. It may be the beginning of a cold or I may suffer from angina (inflammation of the throat). If it is severe angina, I will not go to work. To gain more insight, I can take my temperature, and I can look down my throat for yellow spots.

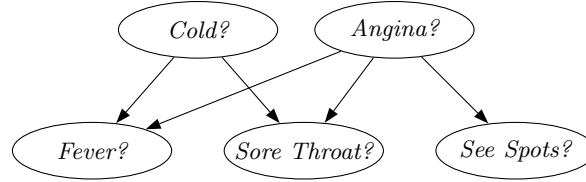
Here we have five hypothesis events *Cold?* {no, yes} and *Angina?* {no, mild, severe}. The hypothesis events must be organized into a set of variables with

mutually exclusive and exhaustive states. We may use the variables indicated previously, but we may also use only one variable *Sick?* with states  $\{no, cold, mild\ angina, severe\ angina\}$ . In the latter case, suffering from both cold and angina is excluded as a possibility. We choose to use the two variables *Cold?* and *Angina?*.

The information variables are *Sore Throat?*  $\{no, yes\}$ , *See Spots?*  $\{no, yes\}$ , and *Fever?*  $\{no, low, high\}$ . The variable *Fever?* causes a problem because it really is continuous. In Section 3.3.8, we give methods on how to deal with continuous variables.

Now it is time to consider the causal structure between the variables. We need not worry about how information is transmitted through the network. The only thing to worry about is which variables have a direct causal impact on other variables.

In this example, we have that *Cold?* has a causal impact on *Sore Throat?* and *Fever?* while *Angina?* has an impact on all information variables. The model is given in Figure 3.6.



**Fig. 3.6.** A model for *Cold* or *angina?*.

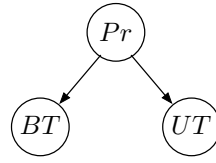
The next thing to check is whether the conditional independences laid down in the model correspond to reality. For example, the model in Figure 3.6 yields that if we know the state of *Angina?*, then seeing spots will not have an impact on the expectation either for *Fever?* or for *Sore Throat?*. If we do not agree, we may introduce a link from *See Spots?* to, for example, *Fever?*. For now, we accept the conditional independences given by the model.

### 3.1.3 Insemination

Six weeks after insemination of a cow, you can perform two tests to determine whether the cow is pregnant: a blood test and a urine test.

Following the method from Section 3.1.1, we construct a model as in Figure 3.7. The variable *Pr*  $\{yes, no\}$  represents a possible pregnancy, and *BT*  $\{pos, neg\}$  and *UT*  $\{pos, neg\}$  represent the result of the blood test and the urine test, respectively.

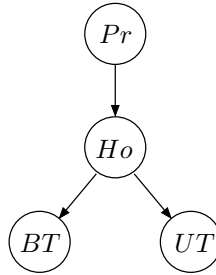
Next, we will analyze the conditional independences stated by the model. We ask the expert whether it is correct that the outcomes of the two tests are independent given *Pr*. More specifically, assume that we know the cow is pregnant. From this, we infer some expectations for the test results. Now, if we



**Fig. 3.7.** A model for pregnancy.

get a negative test result from the blood test, will this change our expectation for the urine test? The experts say that it will, and we must conclude that the model is not a proper reflection of reality.

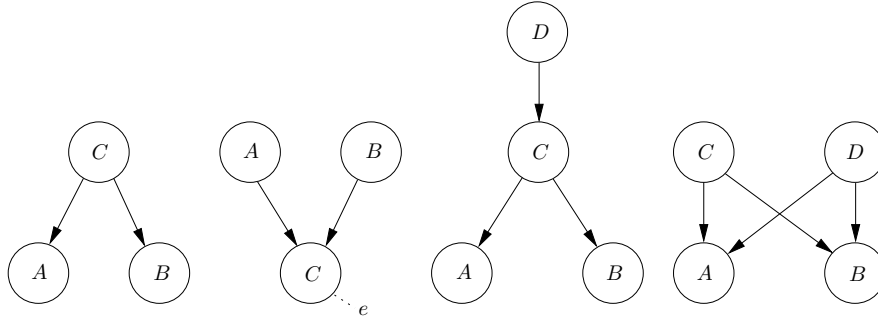
There are several ways to change the model. You might, for example, introduce a link between the two test nodes, but there is no natural direction. To find out what to do, you must study the process more carefully, and it turns out that what the two tests actually do is tracing indications of hormonal changes in the cow. A more refined model will involve a variable  $Ho$ , reflecting whether hormonal changes have taken place in the cow, and the model will be as in Figure 3.8.



**Fig. 3.8.** A more correct model for pregnancy. Both the blood test ( $BT$ ) and the urine test ( $UT$ ) measure the hormonal state ( $Ho$ ).

For the model in Figure 3.8, it does not hold that  $BT$  and  $UT$  are independent given  $Pr$ . The model states that  $BT$  and  $UT$  are independent given  $Ho$  (which should be checked). If the model in Figure 3.7 is used for diagnosing a possible pregnancy, a negative outcome of both the blood test and the urine test will be counted as two independent pieces of evidence and therefore overestimate the probability for the insemination to have failed (see Exercise 3.8).

In the model in Figure 3.8, we have introduced the variable  $Ho$ , which is neither a hypothesis variable nor an information variable. Such variables are called *mediating variables*. Mediating variables are often introduced when two variables are not (conditionally) independent as opposed to the situation in the current model. Some standard situations are illustrated in Fig. 3.9.



**Fig. 3.9.** Examples where an intermediate variable  $C$  “solves” undirected dependencies. In examples (a) and (b),  $A$  and  $B$  are not independent, whereas  $A$  and  $B$  are not independent given  $D$  in examples (c) and (d).

### 3.1.4 A Simplified Poker Game

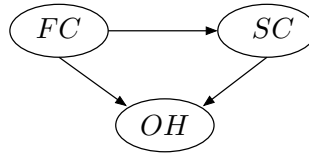
In this poker game, each player receives three cards and is allowed two rounds of changing cards. In the first round, you may discard any number of cards from your hand and get replacements from the pack of cards. In the second round, you may discard at most two cards. After the two rounds of card changing, I am interested in an estimate of my opponent’s hand.

The hypothesis events are the various types of hands in the game. They may be classified in the following way (in increasing rank): nothing special, 1 ace, 2 of the same value, 2 aces, flush (3 of a suit), straight (3 of consecutive value), 3 of the same value, straight flush. Ambiguities are resolved according to rank. This is, of course, a simplification, but it is often necessary to do so when modeling. The hypothesis events are collected into one hypothesis variable  $OH$  (opponent’s hand) with the preceding classes as states.

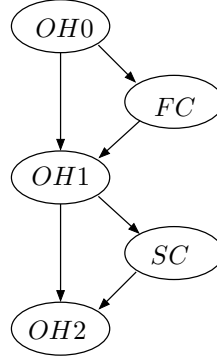
The only information to acquire is the number of cards the player discards in the two rounds. Therefore, the information variables are  $FC$  (first change) with states  $0, 1, 2, 3$  and  $SC$  (second change) with states  $0, 1, 2$ . By saying this, we are making an approximation again. The information on the cards you have seen is relevant for your opponent’s hand. If, for example, you have seen three aces, then he cannot have two aces.

A causal structure for the information variables and the hypothesis variable could be as in Figure 3.10. However, this structure will leave us with no clue as to how to specify the probabilities.

What we need are mediating variables describing the opponent’s hands in the process: the initial hand  $OH0$  and the hand  $OH1$  after the first change of cards. The causal structure will then be as in Figure 3.11.



**Fig. 3.10.** An oversimplified structure for the poker game. The variables are *FC* (first change), *SC* (second change), and *OH* (opponent's hand).



**Fig. 3.11.** A structure for the poker game. The two mediating variables *OH0* and *OH1* are introduced. *OH2* is the variable for my opponent's final hand.

To determine the states of *OH0* and *OH1*, we must produce a classification that is relevant for determining the states of the children (*FC* and *OH1*, say). We may let *OH0* and *OH1* have the states *nothing special*, *1 ace*, *2 of consecutive value*, *2 of a suit*, *2 of the same value*, *2 of a suit and 2 of consecutive value*, *2 of a suit and 2 of the same value*, *2 of consecutive value and 2 of the same value*, *flush*, *straight*, *3 of the same value*, *straight flush*.

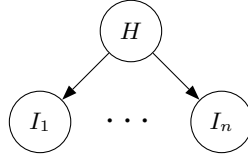
We defer further discussion of the classification to the section on specifying the probabilities (Section 3.2).

### 3.1.5 Naïve Bayes Models

In the previous sections we saw examples of Bayesian networks that were designed to capture the independence properties in the domains being modeled. However, the first Bayesian diagnostic systems were actually constructed based on much simpler models, namely so-called *naïve Bayes models*. In a naïve Bayes model the information variables are assumed to be independent given the hypothesis variable (see Figure 3.12).

Using this assumption the conditional probability distribution for the hypothesis variable given the information variables is very easy to calculate, and the overall process (from model specification to probability updating) can be summarized as follows:





**Fig. 3.12.** A naïve Bayes model.

- Let the possible diseases be collected into one hypothesis variable  $H$  with prior probability  $P(H)$ .
- For all information variables  $I$ , acquire the conditional probability distribution  $P(I | H)$  (the *likelihood* of  $H$  given  $I$ ).
- For any set of observations  $f_1, \dots, f_n$  on the variables  $I_1, \dots, I_n$ , calculate the product  $P(f_1, \dots, f_n | H) = P(f_1 | H) \cdot P(f_2 | H) \cdot \dots \cdot P(f_n | H)$ . This product is also called the likelihood for  $H$  given  $f_1, \dots, f_n$ . The posterior probability for  $H$  is then calculated as:

$$\begin{aligned}
 P(H | f_1, \dots, f_n) &= \mu P(H) P(f_1, \dots, f_n | H) \\
 &= \mu P(H) \prod_{i=1}^n P(f_i | H),
 \end{aligned} \tag{3.1}$$

where  $\mu = 1/P(f_1, \dots, f_n)$  is a normalization constant.

What is particularly attractive with the calculation in Equation 3.1 is that the time complexity is linear in the number of information variables, and that each term in the product only involves two numbers (assuming that the hypothesis variable is binary), one for  $P(f_i | H = y)$  and one for  $P(f_i | H = n)$ . On the other hand, as we also saw from the insemination example, the independence assumption needs not hold, and if the model is used anyway, the conclusions may be misleading. However, in certain application areas (such as diagnosis) the Naïve Bayes model has shown to provide very good performance, even when the independence assumption is violated. This is partly due to the fact that for many diagnostic problems we are only interested in identifying the most probable disease. In other words, if the conditional independence assumption does not change which state has the highest probability, then the naïve Bayes model can be used without affecting the performance of the system. We shall return to these models in Section 8.1.

### 3.1.6 Causality

In the examples presented in the previous section, there was no problem in establishing the links and their directions. However, you cannot expect this part of the modeling always to go smoothly.

First, causal relations are not always obvious – recall the debates on whether smoking causes lung cancer or whether a person’s sex has an impact on his/her ability in the technical sciences. Furthermore, causality is not